

MediAlert - Medication Reminder & Drug Interaction Checker

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1. Introduction

1.1 Problem Statement

Medication non-compliance is one of the most serious yet preventable crises in modern healthcare. According to the **World Health Organization**, failure to take medications as prescribed is responsible for **approximately 125,000 preventable deaths** every year in the United States alone and accounts for 10% of all hospitalizations globally. This problem is most acute among elderly patients and individuals managing multiple chronic conditions simultaneously — a situation clinically referred to as **polypharmacy**, defined as the concurrent use of **five or more medications**.

Patients with polypharmacy face three compounding challenges. First, they frequently **miss scheduled doses** due to forgetfulness or complex pill regimens. Second, they inadvertently **double-dose** because they cannot remember whether they have already taken a medication. Third, and most dangerously, they unknowingly combine medications that **interact with each other** in ways that can cause serious adverse effects — including **stroke, internal bleeding, organ failure, and death**.

Despite the magnitude of this problem, no **accessible, user-friendly, and free consumer tool** currently exists that combines **smart reminder notifications** with **real-time drug interaction checking**. Patients currently rely on paper pill organizers, handwritten notes, or generic calendar alarms — none of which check for dangerous drug interactions. Expensive smartwatch ecosystems offer reminder features but without **pharmacological intelligence**. Pharmacy consultation is inconsistent and unavailable at **2 AM** when a patient considers taking an unfamiliar over-the-counter supplement alongside their prescription medications.

1.2 Proposed Solution

MediAlert is a life-critical, web-based medication management platform designed to solve all three of the above problems in a single, integrated application. Built on **.NET Blazor (WebAssembly)**, MediAlert allows patients to:

- Log their complete **prescription and supplement list** with dosage, frequency, and duration details.
- Receive **smart, scheduled dose reminders** at exact intervals with missed-dose alerts.
- Instantly check whether any new medication **interacts dangerously** with their existing medications — before filling the prescription.
- Allow **family caregivers** to remotely monitor compliance for elderly relatives.
- Export **monthly medication compliance reports** for sharing with healthcare providers.

The platform is entirely web-based, requiring no app installation, and is accessible from any device — smartphone, tablet, or desktop — making it universally accessible to elderly and low-mobility patients.

1.3 Objectives

1. Develop a secure, **multi-user medication management** web application using **.NET Blazor**.
2. Implement a real-time drug interaction checker powered by the **OpenFDA Drug API**.
3. Build a **smart dose scheduling engine** with compliance tracking and missed-dose alerting.
4. Create a caregiver portal enabling remote family monitoring of medication compliance.
5. Deliver **10 full CRUD** modules covering all core aspects of medication management.

6. Design a **polished, accessible UI** with **Figma** mockups for all major screens.

1.4 Domain

Healthcare & Patient Safety — specifically the sub-domain of medication management, adverse drug event prevention, and chronic disease compliance for elderly and polypharmacy patients.

1.5 Target User

User Group	How MediAlert Helps Them
Elderly Patients	Manage complex multi-medication regimens without relying on memory alone; receive timely reminders
Chronic Illness Patients	Safely add new prescriptions while checking for dangerous interactions with existing medications
Family Caregivers	Monitor a parent or relative's medication compliance remotely from any device via approved patient linking
General Users	Track over-the-counter supplements and vitamins alongside prescription medicines
Pharmacists	Quickly reference a patient's full medication history before dispensing
Licensed Doctors	Register a doctor profile, manage patient consultations, conduct video consultations, and track consultation history (Premium feature for patients)

2. Features & CRUD Operations

2.1 Core Feature Overview

- Full prescription log with dosage, schedule, frequency, and **refill tracking**
- Smart dose reminders with **browser notifications** and missed-dose history log
- Real-time drug interaction checker using the **OpenFDA Drug API**
- Caregiver portal for monitoring elderly family members remotely via patient-approved linking
- Appointment scheduler supporting doctor consultations, video consultations, and pharmacy visits
- Refill reminders automatically generated based on medication end dates and lead times — displayed alongside appointments in a unified calendar view
- Monthly health compliance reports exportable as PDF
- **Role-based access control**: Patient, Caregiver, Doctor, and Admin roles
- Doctor Management with profile registration, availability calendar, consultation booking, and **Zoom-based video consultations** (Premium feature)
- Subscription & Payment Management via **Stripe with Free and Premium** (Monthly) tiers; each user may have only one active subscription
- **Admin Panel** for user oversight, doctor verification, and platform governance
- Secure authentication with **email verification** and JWT tokens
- Responsive design accessible on desktop

2.2 CRUD Operations (13 Total)

#	CRUD Operation Description	Module / Entity
01	Create, edit, and delete user accounts and personal health profiles	Auth & User Management
02	Create, edit, and delete medications with full dosage, frequency, and duration details	Medications
03	Create, update, and delete dose schedules and reminder times for each medication	Dose Schedules
04	Log, update, and delete individual dose intake records (taken / skipped / delayed)	Intake Logs
05	Add, update, and remove personal health conditions linked to the user profile	Health Profile
06	Create, edit, and delete caregiver accounts with linking and approval workflow	Caregiver Accounts
07	Save, view, and delete drug interaction check reports generated from API results	Interaction Reports

#	CRUD Operation Description	Module / Entity
08	Set, update, and delete prescription refill reminder dates per medication	Refill Reminders
09	Create, update, and delete scheduled doctor consultations, video consultations, and pharmacy visits	Appointments
10	Generate, view, and delete monthly medication compliance reports with PDF export	Reports
11	Register, update, and manage doctor profiles, availability slots, patient links, and consultations	Doctor Management
12	Create, upgrade, downgrade, and cancel subscriptions (Free / Premium); manage invoices and payment methods. Each user may have only one active subscription.	Subscription & Payment
13	View, suspend/delete users; approve/reject doctors; view/cancel consultations; manage subscriptions	Admin Panel

2.3 API Integrations

API / Service	Purpose in MediAlert	Integration Point
OpenFDA Drug API	Checks for drug-drug interactions; retrieves drug warnings, adverse effects, and label data in real time	Drug Interaction Checker — Module 5
Zoom API	Enables real-time video consultations between patients and doctors; auto-generates meeting links	Doctor Consultations — Module 9
Stripe API	Manages subscription billing for Free and Premium tiers, payment methods, invoice generation, and webhooks	Subscription & Billing — Module 10

3. Application Modules

MediAlert is organized into 11 primary modules, each representing a coherent functional area of the application. All modules share a common authentication layer and database context.

Module	Module Name	Description
Module 1	Auth & User Management	Secure user registration, login, email verification, JWT-based session management, role assignment (Patient / Caregiver / Doctor / Admin), and personal health profile CRUD.
Module 2	Medication Management	Full CRUD for the medication catalog: drug name, dosage, form (tablet/liquid/injection), frequency, start date, end date, prescribing physician, and pharmacy details.
Module 3	Dose Scheduling Engine	Generates individualized dose schedules from medication frequency settings. Calculates exact reminder timestamps, handles multi-dose-per-day schedules, and updates records in real time.
Module 4	Intake & Compliance Logging	Records each dose event (taken, skipped, delayed) with timestamp. Computes compliance percentage per medication and overall. Stores intake history with exportable data.
Module 5	Drug Interaction Checker	Calls OpenFDA Drug API with the user's current medication list plus a newly entered drug. Parses interaction data and presents results in plain language with severity levels (Minor / Moderate / Major / Contraindicated).
Module 6	Caregiver Portal	Patients and caregivers create accounts independently. A patient links a caregiver by searching the caregiver's email and sending a link request. The caregiver approves or rejects the request. After approval, the caregiver gains access to real-time compliance overview, missed doses, upcoming reminders, and read-only access to the patient's medication list.
Module 7	Appointments & Refill Reminders	CRUD for appointments including doctor consultations, video consultations via Zoom, and pharmacy visits. Refill reminders are a separate feature, automatically generated based on prescription end dates and pharmacy lead times. Both appointments and refill reminders appear in a unified calendar-style view.
Module 8	Reports	Generates monthly compliance reports as an in-app dashboard and exportable PDF.
Module 9	Doctor Management & Consultation	Doctor registration with license verification, profile management, availability calendar, patient-doctor linking via doctor discovery, consultation booking, Zoom meeting management with auto-generated links, consultation history tracking, and doctor dashboard analytics. Doctor consultation features are premium-only; patients without a Premium subscription must upgrade before accessing appointment booking or video consultations.

Module	Module Name	Description
Module 10	Subscription & Payment Management	Stripe integration for subscription billing with two tiers. Free tier includes: medication management, dose scheduling, medicine reminders, intake logging, compliance reports, and refill reminders. Premium (Monthly) tier includes all Free features plus: doctor discovery, appointment booking, video consultations, consultation history, and consultation notes. Each user may have only one active subscription at a time. Manages payment method management, invoice history, tier upgrades/downgrades with prorated billing, usage statistics, and automated billing notifications.
Module 11	Admin Panel	User management with suspension/deletion, doctor verification with license approval, subscription oversight with tier adjustments and refunds, and consultation oversight with read-only access and cancellation capability.

4. Figma Mockup Screens

Five complete screen mockups have been designed in Figma representing the primary user journeys within MediAlert. Each screen follows a consistent design system with a blue-and-white medical aesthetic, accessibility-compliant contrast ratios, and responsive layout breakpoints.

Screen 1 — Patient Dashboard

Figure 1 — MediAlert Dashboard Mockup (Screen 1)

- The dashboard is the first screen a logged-in patient sees. It displays:
- Today's scheduled medications in chronological order with a **'Take Dose'** action button beside each entry
- A compliance streak counter showing consecutive days of perfect adherence
- A **ring-chart summary** of today's compliance progress (e.g., 3 of 5 doses taken)
- Quick-access cards for Drug Interaction Checker, Upcoming Appointments, and Refill Alerts
- **Weather alert banner** (shown conditionally when storage conditions are unsafe)
- Navigation sidebar with links to all modules



Good morning, James

Monday, April 27, 2026

TODAY'S DOSES

6

3 taken, 3 pending

COMPLIANCE STREAK

24

days in a row

UPCOMING DOSES

3

next 12 hours

Today's Schedule

Metformin

500 mg • Morning dose

✓ Taken

08:30 AM

Lisinopril

10 mg • With breakfast

✓ Taken

08:45 AM

Atorvastatin

20 mg • Evening dose

Pending

09:00 PM

Amlodipine

5 mg • Daily

Pending

06:00 PM

Aspirin

81 mg • Daily

Pending

07:00 PM

Vitamin D3

2000 IU • Morning

Missed

08:30 AM

Screen 2 — Medication Detail Page

Figure 2 — Medication Detail & Schedule Page (Screen 2)

The medication detail page provides a full view of a single medication entry:

- **Drug name**, dosage form, strength, and prescribing physician
- Dose schedule table showing each day's intake times
- 7-day and 30-day history bar chart showing adherence per dose slot
- Refill countdown progress bar: **'Refill needed in X days'**
- Edit and Delete actions with a confirmation dialog
- Link to run drug interaction check for this specific medication

[← Back to dashboard](#)

Metformin

DOSAGE

500 mg

FREQUENCY

Twice daily

START DATE

Jan 15, 2024

PRESCRIBER

Dr. Smith

REFILL IN

12

days • Next refill: May 9

[Request Refill Early](#)

Intake History (Last 7 Days)

Mon

Tue

Wed

Thu

Fri

Sat

Sun

Scheduled Times



Morning
8:00 AM



Evening
8:00 PM



Screen 3 — Drug Interaction Checker Results

Figure 3 — Drug Interaction Checker Results (Screen 3)

The interaction checker screen allows the user to input a new drug and view results:

- Search field for entering a new drug name with auto-suggest from OpenFDA
- **User's current medication** list displayed as tags for context
- Results card for each detected interaction, showing: drug pair, **severity badge**, plain-language explanation, and **'Consult Your Doctor'** advisory
- Severity badge color system: Blue = Minor, Amber = Moderate, Red = Major, Dark Red = Contraindicated
- **'Save Report'** button to store the check result in the user's Interaction Reports
- Empty state illustration when no interactions are detected ('Your medications appear compatible')

[← Back to medications](#)

Drug Interaction Checker

Add a new medication to check for potential interactions with your current medications

[Cancel Name](#)

Results for "Warfarin"

SEVERE Warfarin + Aspirin

Taking these medications together significantly increases your bleeding risk. Both drugs prevent blood clotting, and combining them can lead to serious bleeding complications.

Recommendation: Do not take these medications together without explicit approval from your doctor. Contact your prescriber immediately.

MODERATE Warfarin + Metformin

Metformin may enhance the anticoagulant effect of Warfarin, increasing your bleeding risk. This combination requires careful monitoring.

Recommendation: Safe to take together with regular INR monitoring. Inform your doctor about this combination.

MILD Warfarin + Vitamin D

No significant interaction expected. Vitamin D supplementation is generally safe with Warfarin at standard doses.

Recommendation: Safe to take together. Monitor your vitamin D levels regularly.

Clear Interactions



No Interactions Found

Metformin does not interact significantly with Vitamin D or Lisinopril

[Save This Report](#)[Share with Doctor](#)

Screen 4 — Caregiver Monitoring Panel

Figure 4 — Caregiver Monitoring Panel (Screen 4)

The caregiver portal gives family members a consolidated view of their linked patient(s):

- Patient selector dropdown for **caregivers** managing multiple family members
- **Compliance overview card**: overall adherence percentage for the past 7 / 14 / 30 days (tab toggle)
- Medication-by-medication compliance row list showing each drug's individual adherence
- Missed dose alert log with timestamps and medication names
- Upcoming reminders section showing the patient's next 5 scheduled doses
- 'Contact Patient' quick-action link (mail to or phone)



MediAlert

Dashboard

My Medications

Family Members

Settings

Preferences

Family Members

Monitor medication compliance for your elderly relatives

Monitoring

Alerts & Notifications

Settings



Missed Dose Alert

Margaret missed her 2:00 PM blood pressure medication. You can message her or contact her doctor.

MJ

Margaret Johnson

Mother • Age 78

TODAY'S DOSES

Morning (8:00 AM)



Afternoon (2:00 PM)



Evening (8:00 PM)



92%

THIS MONTH

18

DAYS STREAK

1

MISSED

View Full History

RJ

Robert Johnson

Father • Age 81

TODAY'S DOSES

Morning (7:30 AM)



Afternoon (1:00 PM)



Evening (7:30 PM)



98%

THIS MONTH

28

DAYS STREAK

0

MISSED

View Full History

EJ

Elizabeth Johnson

Sister • Age 45

Last seen: 3 hours ago

No data available

COMPLIANCE

STREAK

MISSED

View Profile

Screen 5 — Monthly Compliance Report

Figure 5 — Monthly Compliance Report (Screen 5)

The reports screen generates a detailed monthly breakdown of the user's medication adherence:

- Month selector at the top with navigation arrows
- **Summary card:** overall compliance percentage, total doses scheduled, taken, missed, and delayed
- Line chart showing daily compliance rate across the month (x-axis = days, y-axis = % compliance)
- Medication breakdown table: each drug row shows **compliance %**, doses taken, missed, and late
- Most-missed medication highlighted in red with a pattern note (e.g., 'Consistently missed after 10 PM')
- 'Export as PDF' button generating a professionally **formatted PDF report** for sharing with physicians



5. Team Responsibilities

The team consists of 3 members. Responsibilities are distributed to ensure each member handles a balanced mix of frontend (UI/UX) and backend (API, database, business logic) work. Each member leads 2–3 modules.

Member	Primary Modules	Frontend Responsibilities	Backend Responsibilities
Member 1 (Frontend Lead)	Module 1: Auth & User Mgmt Module 5: Drug Interaction Checker Module 6: Caregiver Portal Module 9: Doctor Management Module 11: Admin Panel (partial)	Login/registration UI, user profile forms, drug interaction checker UI, caregiver monitoring panel, doctor registration & availability UI, admin user management & doctor verification UI, Figma mockups for Screens 1, 4 & 6	JWT authentication, OpenFDA API integration, role-based access control (Patient / Caregiver / Doctor / Admin), caregiver-patient link request and approval logic, Zoom API service, doctor-patient linking logic, admin suspension/verification endpoints
Member 2 (Backend Lead)	Module 2: Medication Mgmt Module 3: Dose Scheduling Engine Module 4: Intake & Compliance Logging	Medication CRUD forms, dose schedule editor UI, intake log view with status indicators, history chart (Screen 2), Figma mockups for Screens 2 & 3	Medication database schema & EF Core models, scheduling engine algorithm, intake log processing, compliance percentage calculation, database migrations
Member 3 (Full Stack / Integration)	Module 7: Appointments & Refill Module 8: Reports Module 10: Subscription & Payment Module 11: Admin Panel (partial)	Appointment scheduler UI (doctor consultations, video consultations, pharmacy visits), refill reminder cards, monthly report dashboard, PDF export button, subscription & billing dashboard UI, admin consultation oversight & subscription management UI, Figma mockups for Screens 5 & 7	Appointment CRUD (doctor consultations, video consultations, pharmacy visits), refill automation based on medication end dates and lead times, unified calendar view, PDF generation, Stripe API service, subscription billing (Free and Premium tiers), payment webhooks, admin consultation cancellation & refund endpoints, deployment

All three members jointly contribute to: database schema design, API endpoint definition, code reviews, integration testing, and final Figma design review.

6. Iteration / Sprint Plan

MediAlert is developed across a **4-week sprint cycle**. Each sprint is one week in duration and delivers specific, testable milestones. The sprint plan distributes work across all three team members with clear ownership per deliverable.

Sprint	Duration	Goals & Deliverables	Member 1	Member 2	Member 3
Sprint 1	Week 1	Project setup, DB schema, authentication with role assignment (Patient / Caregiver / Doctor / Admin), user profile CRUD, medication CRUD, base UI layout	Auth UI (login/register), user profile forms, project repo setup	DB schema design, medication CRUD backend, EF Core models & migrations	Base Blazor layout, navigation sidebar, component library, API skeleton
Sprint 2	Week 2	Dose scheduling engine, reminder notifications, intake logging, compliance tracking logic	Intake log UI, compliance dashboard widgets, missed-dose alert display	Scheduling engine algorithm, reminder timestamp generation, intake log backend	Notification service integration, compliance percentage calculator, history data aggregation
Sprint 3	Week 3	OpenFDA API drug interaction checker, interaction results UI, caregiver linking with email search and approval workflow	Drug interaction checker UI, caregiver linking request UI with approval workflow, role-based navigation	Interaction report CRUD, results display with severity badges	OpenFDA API integration service, caregiver-patient link request and approval logic, permission guard middleware
Sprint 4	Week 4	Appointments (doctor consultations, video consultations, pharmacy visits), refill reminders in unified calendar, compliance reports (PDF export), Doctor Management, Subscription & Payment (Free and Premium tiers), Admin Panel, UI polish, Figma finalization	Doctor registration & availability UI, admin user management UI, UI polish, accessibility audit, final Figma finalization	Compliance report data aggregation, chart rendering, medication breakdown table	Appointment CRUD, refill automation, unified calendar view, PDF export, Stripe API (Free/Premium), Zoom API, admin oversight UI, deployment

6.1 Sprint Milestones Summary

- **Sprint 1 Milestone:** Fully functional authentication system with role-based assignment (Patient, Caregiver, Doctor, Admin), user profile management, and medication CRUD with database connectivity.
- **Sprint 2 Milestone:** Operational dose scheduling engine, reminder system, and intake compliance logging with history tracking.
- **Sprint 3 Milestone:** Working OpenFDA drug interaction checker, persistent interaction reports, and caregiver portal with patient-initiated email search, link request, and caregiver approval workflow.
- **Sprint 4 Milestone:** Complete application with all 11 modules integrated, unified appointment and refill calendar (doctor consultations, video consultations, pharmacy visits), PDF reports, subscription billing (Free and Premium tiers), doctor consultation features restricted to Premium subscribers, Zoom video consultations, Admin Panel, fully polished UI, and deployment-ready build.

7. Conclusion

MediAlert addresses a genuine, life-threatening gap in consumer healthcare technology. By combining **smart medication reminders**, **real-time drug interaction intelligence**, and remote caregiver monitoring in a single, free, web-based platform, MediAlert has the potential to prevent thousands of adverse drug events and hospitalizations for its users.

The application is technically grounded: it leverages the **.NET Blazor WebAssembly** framework for a high-performance single-page application experience, integrates two industry-standard external APIs (**OpenFDA**), and implements **10 complete CRUD operation sets** covering every major aspect of medication management. The **4-week sprint plan** is realistic and achievable for a team of three, with clearly defined responsibilities that balance frontend and backend work across all members.

From a course perspective, MediAlert demonstrates mastery of **component-based UI architecture**, **RESTful API consumption**, **database design**, **role-based access control**, third-party API integration, and professional UI/UX design. The healthcare domain requires particular care in **data accuracy** and user experience clarity, both of which have been prioritized throughout the design process.

Most importantly, MediAlert is not merely a course exercise. With appropriate hosting, it could be deployed as a **real service** that directly improves the quality of life for elderly patients, their families, and the broader healthcare system. The team is committed to building something genuinely useful, technically excellent, and professionally delivered.